

The Benefits of Motivational Messaging on Physical Activity

By: 6993

“I attest that upon completion of this exam, I have destroyed the data file and I acknowledge that these data should not be used for any other purposes.” - 6993

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1 Abstract

Background

Physical inactivity is a leading risk factor for chronic diseases, prompting the use of motivational messaging-such as generic or personalized prompts-to increase physical activity. This study examines the effectiveness of three messaging interventions by comparing daily step counts over 14 days, assessing differences in short-term response, long-term trends, and individual variability. Additionally, it explores whether baseline factors and early step data can predict later activity levels.

Methods

The study uses statistical models, tests, and measures in order to evaluate the effects of motivational messages on daily step count over time. Multiple linear regression (MLR) with interaction terms helped determine effects of the treatment arm groups on average daily step count and how these effects vary between activity levels. MLR is also used to predict average daily step count given baseline covariates and daily step count in week 1. Linear mixed models were implemented to determine trajectories of daily step count over time. Statistical tests including Tukey's pairwise test, MCAR test, levene's test, and likelihood ratio test were used to evaluate significance in models/covariates and/or draw conclusions. Statistical measures/evaluation used include k-fold cross validation, RMSE and R^2 to evaluate the accuracy of daily step count prediction models. Finally, missingness in the data is handled by multiple imputation and in some cases linear mixed models (LMM more so for missingness in outcome).

Results

This study found that both standard and tailored motivational messaging significantly increased physical activity compared to no messages, with no meaningful difference between the two approaches. This conclusion remains consistent among the 3 activity levels. The study also finds that the 3 intervention groups have differing 14-day trajectories of daily step count. The predictive model's covariate estimates and respective p-values suggest baseline differences between participant's group and activity level alongside specific days of daily step count in week 1 contribute the most to predicting average daily step count in week 2. Finally, individual level variability contributed less to overall variability as compared to between person variability, and the magnitude of individual-level variability is consistent across the three arms control, standard, and tailored.

Conclusion

Overall, motivational messaging (both standard and tailored) have a significant effect of increasing physical activity as compared to no motivational message, but over time the effect diminishes as daily step count decreases in week 2 (visualized in Appendix figure 1).

2 Introduction

Physical inactivity is a leading risk factor for chronic diseases such as "heart disease, type 2 diabetes, and certain cancers," contributing to significant global morbidity and mortality [1]. With health programs attempting to find methods to combat physical inactivity, health technologies such as active video games, health wearable devices, and notably mobile device applications have become more prevalent with the rise of technology in everyday life [2].

Among aforementioned strategies, motivational messaging has shown promise in increasing physical activity. Generic messages (e.g., "Take a walk today!") are simple to implement and can be effective for broad audiences. However, individuals vary in their habits, readiness to change, and responsiveness to feedback, raising the possibility that tailored interventions-which incorporate personal data such as past step counts or demographic profiles-may yield stronger results. By increasing message relevance, fostering self-efficacy, and reducing information overload, tailored content could enhance engagement and long-term behavior change [3].

To address which type of motivation message, if any, are effective in increasing physical activity, a randomized trial was created in which participants were assigned to one of three intervention groups. Participants were observed over 14 days with the amount of steps participants walked being recorded each day in addition to covariates such as age, gender, and activity level written at the start of the trial to check for potential effect on the amount of steps walked daily. The primary objectives of this analysis are to answer the following questions of interest:

1. How do the three intervention groups compare in average daily step count during the first week? How does this comparison vary by baseline activity level?
2. How do the three intervention groups compare in the 14-day trajectories of daily step count?
3. Using each participant's baseline covariates (baseline activity level, age, sex), intervention group, and their daily step counts on Day 1, Day 2, ..., and Day 7, how well can you predict their average daily step count averaged over Days 8-14?
4. To what extent do individuals differ in their daily step count over the course of the 14-day study? Does the magnitude of this individual-level variability differ between the three arms?

3 Data Description

The data (visualized in appendix figure 1) includes 300 participants randomly assigned to one of the three intervention groups via stratified random sampling: control group (0 messages), standard intervention group (one generic message each morning), and tailored intervention group (one personalized message each morning). The measurement of steps taken were collected daily and the longitudinal outcome measured at each visit is:

- Stepcount: Amount of steps taken in a particular day

With other covariates collected including:

- Baseline Activity level (Factor with level = 1 if not active, = 2 if somewhat active, and = 3 if very active)
- Baseline Age (in years)
- Sex (female = 1, male = 0)
- Day (study day from 1 to 14)
- Dow (day of the week represented as Mon, Tue, etc.)
- Group (Factor with Control, Standard, and Tailored)

4 Methods

Missing Data

The trial contains missing data within the covariate Female with 2% of subjects (Appendix Figure 2). Although covariate Female appears to be the only variable with missing values, closer inspection of the outcome variable stepcount reveals several instances of 0 stepcount, meaning not a single step is taken on that particular day. As such, we assume it is impossible to take no steps within a single day and consider them missing values, thus 35% of subjects (Appendix Figure 3) have at least one instance of 0 stepcount. A majority of these instances are intermittent missingness with subjects having 1-2 missing days, but there are a couple instances of monotone missingness and 2 instances of subjects having 0 daily step count for the entire study. In regards to the subjects with 14 days of 0 step count, we remove them from the sample as they do not have any analyzable data for the outcome variable of interest.

A test for missing completely at random (MCAR) from package naniar resulted in a p-value < 0.001 thus rejecting MCAR for our missingness. Logistic regression modeling with the relation between missingness in stepcount (0 = non-zero step count, 1 = 0 step count) and covariates (Group, Act, Age, Female, dow) after converting any 0 value step count instance to NA resulted in no statistically significant predictors indicating no clear covariate causing missingness. Therefore, the data can be either missing at random or not missing at random. With this conclusion, we assume our data is MAR to continue the

analysis and we implement multiple imputation ($m = 5$ imputations) for the analysis in order to handle the missingness within our gender covariate (female) as well as the many instances of 0 in our outcome variable stepcount.

SQ 1

Question 1 part 1 asks between groups control, standard, and tailored, does the average daily step counts in week 1 differ. Question 1 part 2 builds on part 1 to determine whether any of the pairwise comparisons from part 1 remain consistent when accounting for activity level (i.e. whether the conclusions hold within subgroups of similar activity levels).

To solve question 1 part 1, we first subset the data to weeks 1 to 7 as per the question and implement multiple imputation using the mice function (via package mice) with $m = 5$ imputations to simulate the missing data. We then fit the imputation data on a multiple linear regression model, as the step count over week 1 maintains a linear pattern (Appendix Figure 4) and missingness of both gender and stepcount are covered by multiple imputation, in the model below:

$$\begin{aligned} \log(\text{stepcount} + 1) = & \beta_0 + \beta_1 I(\text{Group} = \text{Standard}) + \beta_2 I(\text{Group} = \text{Tailored}) \\ & + \beta_3 I(\text{Act} = \text{somewhat_active}) + \beta_4 I(\text{Act} = \text{very_active}) \\ & + \beta_5 I(\text{Group} = \text{Standard}) \times I(\text{Act} = \text{somewhat_active}) \\ & + \beta_6 I(\text{Group} = \text{Tailored}) \times I(\text{Act} = \text{somewhat_active}) \\ & + \beta_7 I(\text{Group} = \text{Standard}) \times I(\text{Act} = \text{very_active}) \\ & + \beta_8 I(\text{Group} = \text{Tailored}) \times I(\text{Act} = \text{very_active}) + \epsilon_{ij} \end{aligned}$$

Model 1: Model fit for Scientific Question 1

The model appears approximately normal (Appendix figure 5) via qqplot/qqline, but our outcome variable is log transformed by $\log(\text{stepcount} + 1)$ to deal with residuals spreading out for large x values (Appendix figure 6) shown in residual plots. This is calculated using fixed effects group and act as well as interaction between group and act with no other covariates added to focus on only covariates of interest for question 1. We conduct pairwise Tukey's test (using pairs function (graphics package) and emmeans function (emmeans package) to obtain marginal means for pairs function) using model 1 to determine whether the average step count in week 1 varies between the 3 groups control, standard, and tailored with null hypotheses $H_{01}: \mu_{\text{control}} = \mu_{\text{standard}}$, $H_{02}: \mu_{\text{control}} = \mu_{\text{tailored}}$, and $H_{03}: \mu_{\text{standard}} = \mu_{\text{tailored}}$.

A significant result in the p-value of any pairwise test means the daily average step count in week 1 is different between the respective test's 2 groups.

As for question 1 part 2, we again conduct pairwise Tukey's test using model 1 to determine whether the results in part 1 hold conditioned on act. As such, we test null hypotheses H_{01} , H_{02} , and H_{03} for each activity level: not active, somewhat active, and very active. If there is a change in p-value conclusion with hypotheses H_{01} , H_{02} , and H_{03} between the 3 activity levels, then we can conclude that the comparison between groups does vary by activity level. But, if hypotheses H_{01} , H_{02} , and H_{03} are constant between the 3 activity levels, then comparison between groups does not vary by activity level.

SQ 2

Question 2 asks whether the 3 intervention groups have the same rate of daily step count over the course of the 14 day trial.

To compare the 14-day step count trajectories across the three intervention groups, I employed linear mixed effects modeling (lmer function via lme4 package) because it accounts for individual variability in baseline and trajectories, supports non linear trends, we have longitudinal data with daily step count measurements taken repeatedly (represented by days) on the subjects, and integrates well with multiple imputation, which is employed in this problem to account for missing data. The analysis was conducted on log transformed step counts (using $\log(\text{stepcount} + 1)$) to address right skewness while preserving "0" values (if any to prevent $\log(0)$ scenario) and to deal with fanning data within the residual

plot (Appendix figure 7). The primary model included fixed effects for group, day, and day² (to capture nonlinear trajectories shown in days 1-14 via Appendix figure 8), along with their interaction terms (Group*day and Group*day²) to test whether trajectories differed by intervention group. Covariates for activity level, age, and gender were included as additional fixed effects to control for potential confounding. Random effects for each subject allowed individual variability in baseline step counts (intercept) and both linear (day) and quadratic (day²) trajectory components. The data provided to create this model is from the multiple imputation data (with and mice function like with SB1). This model can be shown below:

$$\begin{aligned} \log(\text{stepcount}_{ij} + 1) = & \beta_0 + \beta_1 I(\text{Group}_i = \text{Standard}) + \beta_2 I(\text{Group}_i = \text{Tailored}) \\ & + \beta_3 \text{day}_{ij} + \beta_4 \text{day}_{ij}^2 \\ & + \beta_5 I(\text{Act}_i = \text{somewhat_active}) + \beta_6 I(\text{Act}_i = \text{very_active}) \\ & + \beta_7 \text{Age}_i + \beta_8 I(\text{Female}_i = \text{Female}) \\ & + \beta_9 I(\text{Group}_i = \text{Standard}) \times \text{day}_{ij} \\ & + \beta_{10} I(\text{Group}_i = \text{Tailored}) \times \text{day}_{ij} \\ & + \beta_{11} I(\text{Group}_i = \text{Standard}) \times \text{day}_{ij}^2 \\ & + \beta_{12} I(\text{Group}_i = \text{Tailored}) \times \text{day}_{ij}^2 \\ & + b_{0i} + b_{1i} \text{day}_{ij} + b_{2i} \text{day}_{ij}^2 + \epsilon_{ij} \end{aligned}$$

Model 2: Model fit for Scientific Question 2

Where day_{ij} for example represents day j for subject i. To formally test whether the trajectories differed by group, I compared two models using likelihood ratio test through Rubin's rules (pool.compare function from package mice) to take the imputed data and pool them together to compare a full model containing the Group*day and Group*day² interaction terms vs a reduced model without these interactions. The pool.compare() function provided a test of whether the interaction terms collectively improved model fit thus testing null hypothesis $H_0: \beta_9 = \beta_{10} = \beta_{11} = \beta_{12} = 0$. A significant p-value ($p < 0.05$) would represent that the full model with the interaction terms is a better model fit than without interactions and therefore the intervention groups change in trajectory over time in terms of daily step count. I then examine the individual interaction coefficients from the pooled full model (function pool from package mice) to determine the nature of these differences. This also helps to confirm my results for the previous test via individual p-values for $\beta_9, \beta_{10}, \beta_{11}$, and β_{12} to see if separately they are significant to the outcome daily step count. A significant p-value ($p < 0.05$) with an individual covariate would mean they are significant to the model fit. This two-stage approach - first testing for overall group differences in trajectories, then examining specific interaction terms - provided a comprehensive understanding of how the interventions influence step count patterns over time while properly accounting for the uncertainty introduced by missing data and between-subject variability.

SQ 3

Question 3 specifically asks how accurately can average daily step count over the 2nd week be predicted using baseline covariates (baseline activity level, age, gender), group, and daily step count in the 1st week.

To predict participants' average daily step counts during Days 8-14 using baseline characteristics, group, and daily step counts from days 1-7, the imputed dataset was changed from long to wide format, creating separate columns for daily step counts for all days while preserving baseline covariates (activity level, age, and gender) and intervention group. The target variable - average steps from Days 8-14 - was calculated for each participant, with incomplete cases excluded from analysis. For each imputed dataset, multiple linear regression (MLR) was incorporated instead of linear mixed effects model like with SQ 2. The reasoning is that we are predicting a singular summary measure (week 2 average) for each subject rather than modeling trajectory over time, the relationship between each predictor and outcome should be linear, the data appears normally distributed (appendix figure 9), and the variance of residuals seem to be constant for fitted values (appendix figure 10) which also shows why we do not log transform here

compared to previous SQs. The linear model incorporates both baseline predictors and daily step counts from the first week (Days 1-7) as explanatory variables to help predict the outcome variable average daily steps for week 2 as asked for in the original question. This is shown in the model below (note there is no jth variable like with SQ2 as each individual day is represented by their own covariate):

$$\begin{aligned} \text{avg_week2_steps}_i = & \beta_0 + \beta_1 I(\text{Group}_i = \text{Standard}) + \beta_2 I(\text{Group}_i = \text{Tailored}) \\ & + \beta_3 I(\text{Act}_i = \text{somewhat_active}) + \beta_4 I(\text{Act}_i = \text{very_active}) \\ & + \beta_5 \text{Age}_i + \beta_6 I(\text{Female}_i = \text{Female}) \\ & + \beta_7 \text{day1}_i + \beta_8 \text{day2}_i + \beta_9 \text{day3}_i \\ & + \beta_{10} \text{day4}_i + \beta_{11} \text{day5}_i + \beta_{12} \text{day6}_i + \beta_{13} \text{day7}_i + \epsilon_i \end{aligned}$$

Model 3: Model fit for Scientific Question 3

Where day2_i for example represents daily step count on day 2 for subject i . The analysis employed Rubin's rules (pool function from package mice) to pool regression coefficients across all imputations, generating estimates with appropriate confidence intervals and p-values. Predictive performance was thoroughly evaluated using multiple metrics including R^2 (default R package) which quantifies explained variance in the outcome and RMSE (root mean squared error also default R package) expressed prediction accuracy in interpretable step count units. To ensure robust performance estimates, 5-fold cross-validation is implemented, which involves repeatedly fitting our model on training subsets, and evaluating them on held-out test data (via predict function, default R package). Results of these predictions were then combined and averaged out, visualized through scatterplot comparing observed versus predicted step counts, stratified by intervention group. This approach measures how well early step counts predict future activity levels through statistical metrics (like pooled R^2 values) and meaningful measures (such as step count prediction errors via RMSE), ultimately offering a complete picture of prediction quality while accounting for the uncertainty introduced by missing observations.

SQ 4

Question 4 part 1 asks to obtain and interpret how much participants individually vary in their daily step count over the 14 day trial. Question 4 part 2 on the other hand builds off part 1 to ask if participants being stratified into one of the 3 groups resulted in difference in their variability - specifically, whether the spread or variance of daily step count variability over the 14-day trial varies by intervention group.

To address the question of how individuals differ in their daily step counts over the 14-day study, the analysis combines descriptive statistics and variance decomposition. Descriptive statistics is used to quantify and easily understand the variability in daily step count throughout the study. Meanwhile, variance partitioning is implemented to help determine how much within individual variability actually impacts the overall variability in daily step count which contains within an individual (changes within a single person over time) and between individuals (individual differences in gender, activity level, etc.) variability. To see if the magnitude of this individual level variability differs between the 3 arms control, standard, and tailored, we use descriptive statistics and levene's test. More specifically, we generate descriptive statistics including mean, standard deviation, and range of within individual variability for each group control standard, and tailored to give easily understandable metrics of overall fluctuations between each group over time. Levene's test, (done via leveneTest function in package car) which tests if the variance or magnitude of within person variability differs between groups, typically cannot be used for time data. But, it is applicable here because the 14 days of daily step count variability is converted into a single summary statistic (via variance value of within person variability) that captures how variable subjects are over the trial time. Levene's test is used to check the null hypothesis:

$$H_0: \sigma_{control}^2 = \sigma_{standard}^2 = \sigma_{tailored}^2 \quad \text{where } \sigma_{control}^2 \text{ for example represents variance within individual}$$

variability for group control, and compares variance of within-person variability between the 3 groups. A significant p-value would mean that at least one of the 3 groups has a different magnitude of individual-level variability thus rejecting the null hypothesis and proving that the magnitude of individual level variability differs between the 3 arms.

5 Results

5.1 Exploratory Data Analysis

On top of the missing data explained in methods, one other modification was made to the original data set which is factoring female (gender), dow (day of the week), Group, and Act to be certain they are treated as categorical data.

Once all adjustments have been made to the initial dataset, Table 1 shows the numerical summary of relevant covariates stratified by treatment group. The baseline characteristics of participants across the three treatment groups (control, standard, and tailored) were well-balanced, with no major imbalances, indicating successful randomization. Activity levels showed similar distributions across groups, with the not active category representing the largest proportion in all three groups (53% control, 48% standard, 54% tailored), while somewhat active and very active categories showed comparable proportions ranging from 29-30% and 16-22% respectively across groups. Mean age was consistent across treatment arms, with all groups centering around 28 years, suggesting appropriate age balance in the randomization. The proportion of females within all groups was consistent too with around 40% females per group. Finally, step count data showed substantial variability within all groups, with standard deviations exceeding the means (control: 2621.05 ± 1777.07 , standard: 3881.36 ± 2282.97 , tailored: 3673.87 ± 2454.76), though the tailored and standard groups showed notably higher mean step counts compared to control group.

The spaghetti plot or figure 1 shows substantial variability in step count patterns across all three treatment groups throughout the study period. Participants in the tailored group appear to have more individuals maintaining consistently higher step counts, while the control group shows greater concentration in lower step count ranges, with the standard group demonstrating intermediate patterns. Individual trajectories exhibit considerable day-to-day fluctuation rather than systematic increases over time.

	Control 1400	Standard 1400	Tailored 1400
n			
Act (%)			
not_active	742 (53.0)	672 (48.0)	756 (54.0)
somewhat_active	406 (29.0)	420 (30.0)	420 (30.0)
very_active	252 (18.0)	308 (22.0)	224 (16.0)
Age (mean (SD))	28.39 (5.69)	27.59 (5.63)	27.78 (5.52)
Female = Female (%)	560 (40.0)	518 (38.9)	560 (40.4)
stepcount (mean (SD))	2621.05 (1777.07)	3881.36 (2282.97)	3673.87 (2454.76)

Table 1: Summary of Data by Dose Group

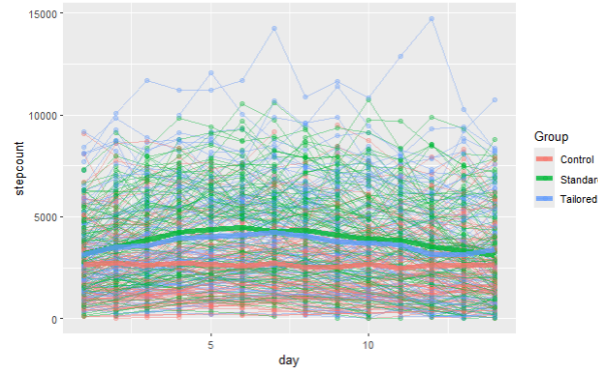


Figure 1: Spaghetti plot for Step Count by Group

5.2 Scientific Question 1

Answering part 1, the analysis compares average daily step counts during the first week across three intervention groups via pairwise comparison test using tukey's test. Table 2 (back transformed outcome for interpretation convenience) shows both the standard and tailored interventions resulted in significantly higher step counts than the control group (standard: 1223 more average steps per day than control, $p < 0.001$; tailored: 1192 more steps per day than control, $p < 0.001$) thus rejecting H_{01} and H_{02} . However, there was no significant difference between the standard and tailored groups (30 steps per day difference, $p = 0.966$), indicating similar effectiveness between these two interventions in the short term thus we cannot reject H_{03} . These results suggest that while both active interventions increased physical activity compared to the control condition, the additional tailoring did not provide further benefits in the first week. The findings raise questions about whether personalization effects might emerge over a longer timeframe, warranting further investigation.

Answering part 2 using table 3, for not active individuals, both standard and tailored interventions significantly increased steps compared to control (standard: 630 more steps per day compared to control, $p < 0.001$; tailored: 684 more steps per day compared to control, $p < 0.001$), with no difference between interventions (54 steps per day difference, $p = 0.761$). Among somewhat active participants, the standard and tailored interventions continued to have significantly increased steps as compared to control (standard: 2111 more steps vs control, $p < 0.001$; tailored: 1527 more steps vs control, $p < 0.001$) and again standard and tailored had close but still not significant difference between each other (583 steps per day difference, $p < 0.061$). For very active individuals, standard and tailored interventions improved steps versus control (standard: 1096 more per day vs control, $p = 0.006$; tailored: 1510 more per day vs control, $p < 0.001$), though neither had a significant difference over the other (414 step difference, $p = 0.59$). These findings conclude that our results we found in part 1 are consistent amongst all activity levels, hence this comparison does not vary by baseline activity level. But, we do notice that although not concluded statistically significant, for somewhat active, standard does noticeably have more average daily step count in week 1 (583 step difference) compared to tailored and vice versa for very active (414 step difference) but nearly identical for not active (54 step difference) which could warrant further looking into and is visualized in appendix figure 11.

Term	Estimate	95% CI	P-value
Control - Standard	-1222.781	(-1423.119, -1022.443)	<0.001
Control - Tailored	-1192.306	(-1400.687, -983.926)	<0.001
Standard - Tailored	30.475	(-207.902, 268.852)	0.966

Table 2: Pairwise Comparison Between Groups

Term	Act	Estimate	95% CI	P-value
Control - Standard	not_active	-630.138	(-755.807, -504.468)	<0.001
Control - Tailored	not_active	-684.135	(-808.685, -559.585)	<0.001
Standard - Tailored	not_active	-53.997	(-204.281, 96.287)	0.761
Control - Standard	somewhat_active	-2111.254	(-2553.806, -1668.702)	<0.001
Control - Tailored	somewhat_active	-1527.663	(-1932.81, -1122.516)	<0.001
Standard - Tailored	somewhat_active	583.592	(78.09, 1089.093)	0.061
Control - Standard	very_active	-1096.173	(-1793.137, -399.208)	0.006
Control - Tailored	very_active	-1510.413	(-2305.751, -715.074)	<0.001
Standard - Tailored	very_active	-414.240	(-1244.344, 415.864)	0.590

Table 3: Pairwise Comparisons Between Groups Conditioned on Activity Level

5.3 Scientific Question 2

The analysis of 14 day step count trajectories reveals key findings about the intervention groups. First, the likelihood ratio test ($p = < 0.001$) provides evidence that step count patterns over time differ significantly between groups, confirming the importance of including group*time interactions in the model. The full model's summary statistics (table 4) also shows these interactions with the standard intervention led to a 13.9% faster daily increase in step counts initially compared to control, but this growth rate slowed by 1.1% per day. The tailored group showed a more moderate 9.4% daily increase, with less deceleration (0.8% per day). Both interventions outperformed control, but the standard group's advantage diminished faster over time. These estimates also support the likelihood ratio test's conclusion in which interaction between standard*day, tailored*day, standard*day², and tailored*day² all produced $p = < 0.001$ thereby each of their respective interactions impact trajectories of daily step count over the course of the experiment.

Term	Estimate (95% CI)	p-value	Very Active	1.253 (1.122, 1.385)	<0.001
(Intercept)	8.126 (7.845, 8.406)	<0.001	Age (years)	-0.034 (-0.043, -0.025)	<0.001
Standard	0.043 (-0.120, 0.205)	0.607	Female	-0.180 (-0.273, -0.088)	<0.001
Tailored	0.152 (-0.008, 0.311)	0.062	Standard:Day	0.139 (0.098, 0.180)	<0.001
Day	0.004 (-0.024, 0.032)	0.788	Tailored:Day	0.094 (0.054, 0.135)	<0.001
Day ²	-0.000 (-0.002, 0.002)	0.682	Standard:Day ²	-0.011 (-0.014, -0.008)	<0.001
Somewhat Active	0.968 (0.856, 1.079)	<0.001	Tailored:Day ²	-0.008 (-0.011, -0.005)	<0.001

Table 4: Summary Statistics of LMM Full Model in Log Form

5.4 Scientific Question 3

The predictive model performed well, explaining 87.7% of variance ($R^2 = 0.877$) in participants' average step counts during days 8-14, with a reasonable prediction error of 735 steps (visualization shown in appendix figure 12). Unusually though, table 5 shows both intervention groups (standard and tailored) were associated with significantly lower future step counts compared to control (414 and 338 fewer steps/day respectively). While baseline activity level showed trends (very active participants tended toward higher future steps compared to not active), only specific days in the first week emerged as significant predictors - particularly days 3, 6, and especially day 7. Age and gender didn't contribute meaningfully to predictions. As such, specific initial baseline variables such as group and activity level alongside specific days of daily step count over week 1 contributed the most to predicting average week 2

daily step count. The conclusions of non-control groups having lower predicted average steps as well as the amount of non-significant predictors (especially for the day variables) for the data though seem unusual for the high predictive value and warrant further investigation.

Term	Estimate (95% CI)	p-value			
(Intercept)	426.54 (-198.68, 1051.75)	0.179	Day_1	-0.11 (-0.22, 0.01)	0.076
Standard	-414.26 (-656.49, -172.03)	<0.001	Day_2	-0.02 (-0.15, 0.12)	0.779
Tailored	-338.17 (-577.52, -98.82)	0.006	Day_3	0.25 (0.09, 0.41)	0.003
Somewhat Active	-295.62 (-596.67, 5.44)	0.054	Day_4	-0.00 (-0.18, 0.18)	0.992
Very Active	388.09 (-46.71, 822.89)	0.080	Day_5	0.10 (-0.10, 0.30)	0.320
Age (years)	-2.35 (-20.40, 15.69)	0.797	Day_6	0.23 (0.09, 0.37)	0.001
Female	-102.40 (-287.12, 82.32)	0.276	Day_7	0.41 (0.30, 0.52)	<0.001

Table 5: Summary Statistics of Predictive Model

5.5 Scientific Question 4

The results in table 6 reveal key insights about step count variability during the 14-day study. The summary statistics which uses multiple imputation data show participants had an average daily step count variability of 867 steps with SD 454 steps, indicating substantial within-person differences (visualization shown in appendix figure 13). More importantly, the variance decomposition shown in table 7 demonstrates that between-person differences (82.5% of total variability) dominated over within-person day-to-day fluctuations (19.1%). This means participants' average step counts varied considerably more from each other than their own daily step count fluctuation across the 14 days. As a practical example, the difference in average step counts variation between a highly active participant and a not active participant (between-person variation) or between a control participant and standard treatment arm participant was typically much larger than the natural day-to-day variations experienced by any single individual.

The second part asks whether the spread of variability in daily step count differs between the 3 arms. Via the levene test, we conclude with a p-value of 0.25 meaning that we cannot reject the null hypothesis stated in SQ 4 and conclude that regardless of treatment arm all participants had around the same magnitude of individual-level variability over the course of the 14-day study. This is further evidenced by table 8 in which while the standard deviations appear somewhat different ranging from 367 to 453, they are modest in comparison to the mean daily step count variability ranging from 591 to 1018 (visualization shown in appendix figure 14).

Count	Mean	SD	Min	Max	Range
300	867.5638	454.1048	65.35133	2536.414	2471.063

Table 6: Summary Statistics for Within Person Variability

Component	Variance	Percentage
Between-person	4146961.3	82.54166
Within-person	958190.8	19.07195
Total	5024082.9	100.00000

Table 7: Variability Impact of Between vs Within Person Variability

Group	Count	Mean	SD	Min	Max	Range
Control	100	591.4039	367.9798	65.35133	2016.534	1951.183
Standard	100	993.0236	406.8927	372.62703	2243.614	1870.987
Tailored	100	1018.2639	453.8225	236.24708	2536.414	2300.167

Table 8: Summary Statistics for Within-Person Variability Stratified by Group

6 Discussion

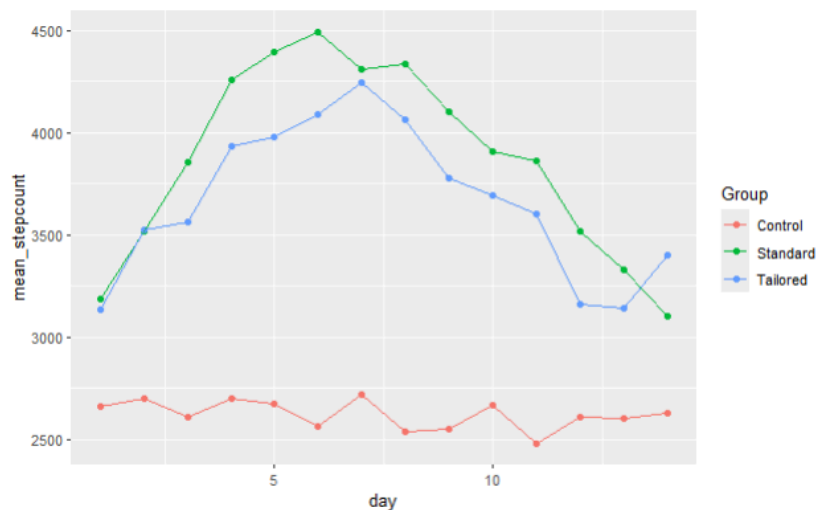
6.1 Conclusion

This study evaluated whether motivational messaging increases physical activity by analyzing daily step counts across three groups: standard messages, tailored messages, and no messages (control). The results demonstrate that both types of motivational messaging significantly increased physical activity compared to the control group, with no significant difference between standard and tailored approaches. These conclusions were also consistent regardless of the baseline activity level. Analysis of 14-day trajectories revealed patterns including the standard group showed an initial 13.9% faster daily increase in steps than control, but this advantage decayed by 1.1% per day. The tailored group exhibited a more moderate but sustained increase (9.4% daily growth, slowing by 0.8% per day). These patterns were statistically significant (all $p < 0.001$), confirming that motivational messaging influences 14-day trajectories over time. Notably, the predictive model yielded unexpected findings. Despite explaining 87.7% of variance in days 8-14 step counts, it projected lower future activity for intervention groups than control. Initial factors such as baseline activity level or arm group as well as daily step count on days during week 1 had the most significance in determining week 2's average daily step count. Variability analysis showed that 82.5% of total variation came from between-person differences (e.g., baseline activity, treatment group) rather than day-to-day fluctuations (19.1%). All groups exhibited similar variability magnitudes (Levene's $p=0.25$), indicating messaging type didn't affect individual variability in step count. These findings highlight that motivational messaging effectively increases activity, and that long-term predictive patterns and the dominance of between-person differences show future interventions should account for individual characteristics. The counterintuitive predictive results warrant further investigation to optimize messaging strategies to maximize physical activity.

6.2 Limitations

While the analyses helped gain valuable insight on motivational messages' impact on physical activity through daily step count, there are several limitations. First, we observed instances of extremely low step counts (potentially representing not wearing the device tracking step count or data errors) that were retained in analyses but may bias results if they are outliers. The missing data mechanism was assumed to be missing at random (MAR), though unmeasured factors affecting both missingness and step counts could violate this assumption. Our analysis was also limited by the available covariates where unmeasured confounders with examples being weather [4] or varying work schedules [5] may influence step patterns but weren't accounted for. We also observe an unexplained significant drop in step counts during the second week across non control groups (shown in appendix figure 1), suggesting possible lessening effect of the messages or other external factors not captured in our model. Additionally, while our multiple linear regression model showed good fit to predicting average step count in week 2, alternative approaches (nonparametric methods, different transformations, etc.) might better capture complex relationships between covariates and step counts. These limitations highlight opportunities for more sophisticated modeling and expanded data collection in future research.

7 Appendix



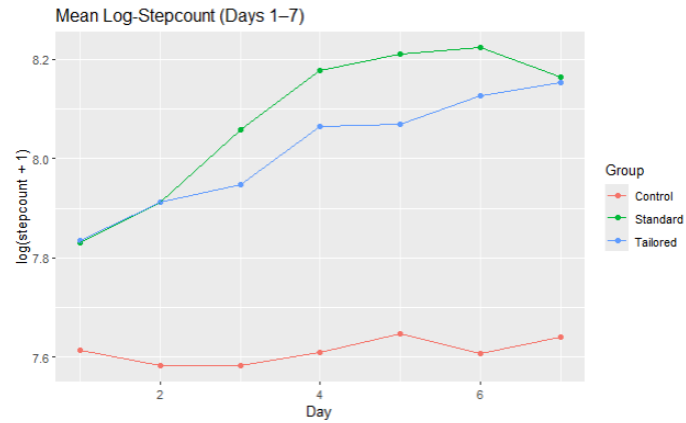
Appendix Figure 1: Mean Step Count Stratified by Group

Subject ID	Treatment Group	Age (years)	Baseline Steps
38	Standard	30	not_active
206	Tailored	22	very_active
237	Standard	25	not_active
252	Standard	23	not_active
281	Standard	36	somewhat_active
283	Standard	35	not_active

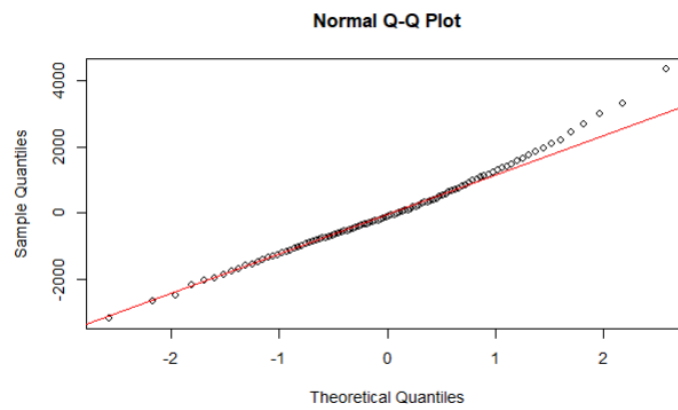
Appendix Figure 2: Missingness within Covariate Female

Missing Entries	Subject IDs	Number of Subjects
1	7, 9, 10, 13, 16, 19, 21, 24, 29, 31, 33, 36, 45, 53, 54, 55, 62, 64, 69, 75, 84, 85, 96, 101, 102, 105, 117, 122, 127, 130, 133, 136, 140, 142, 149, 151, 159, 160, 166, 168, 170, 179, 187, 198, 210, 211, 213, 218, 219, 220, 222, 230, 237, 239, 243, 247, 250, 251, 258, 263, 264, 267, 278, 287, 291, 297, 298, 299, 300	69
2	22, 27, 30, 32, 52, 76, 97, 123, 141, 145, 197, 215, 241, 252, 257, 269, 275	17
3	1, 112, 125, 174, 177, 178, 227, 273	8
4	17, 43, 44, 77, 147, 182, 207	7
6	225	1
9	93	1
14	47, 80	2

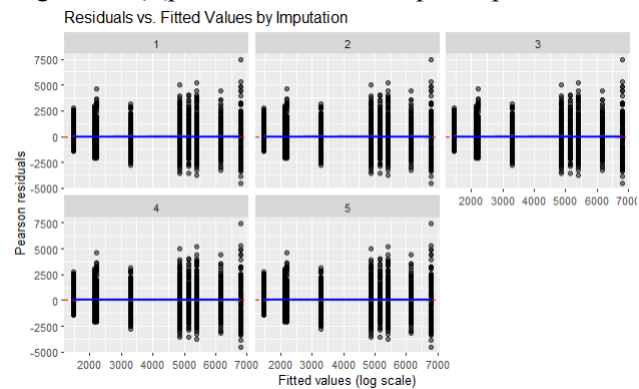
Appendix Figure 3: Missingness within Outcome Variable Stepcount



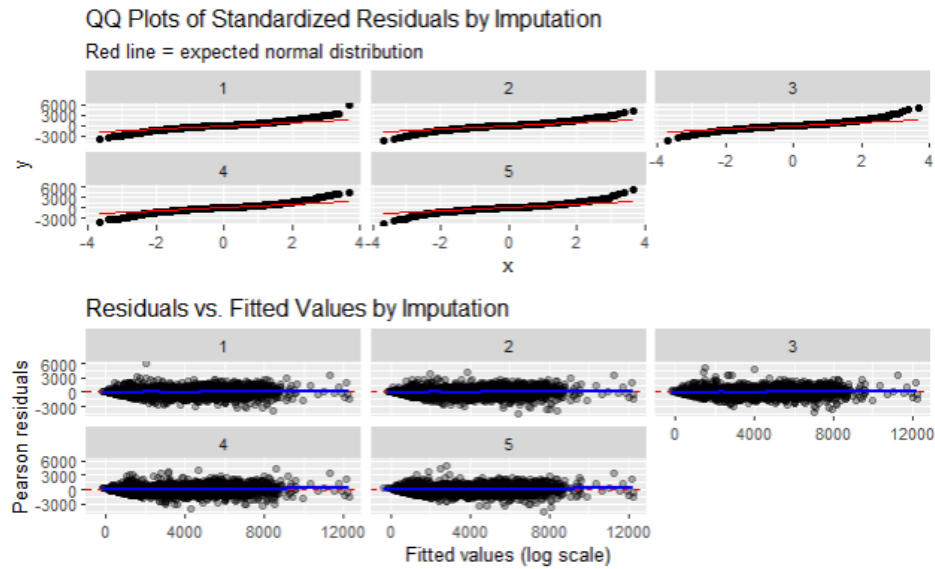
Appendix Figure 4: Log transformed Step Count by Message Group in Week 1



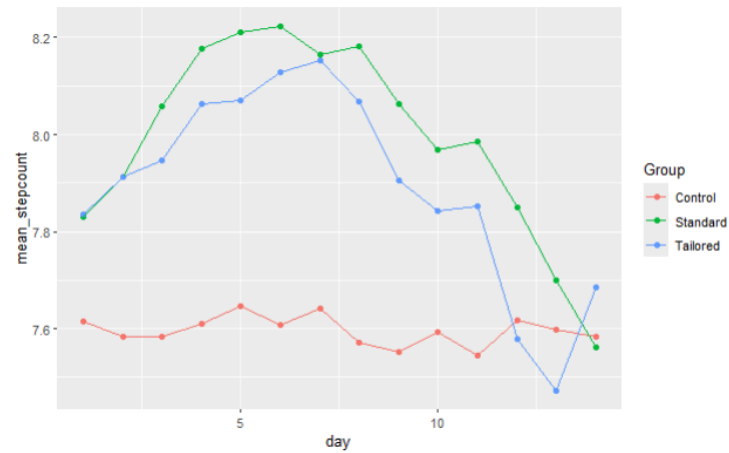
Appendix Figure 5: QQplot for Pooled Multiple Imputation Data for SQ 1



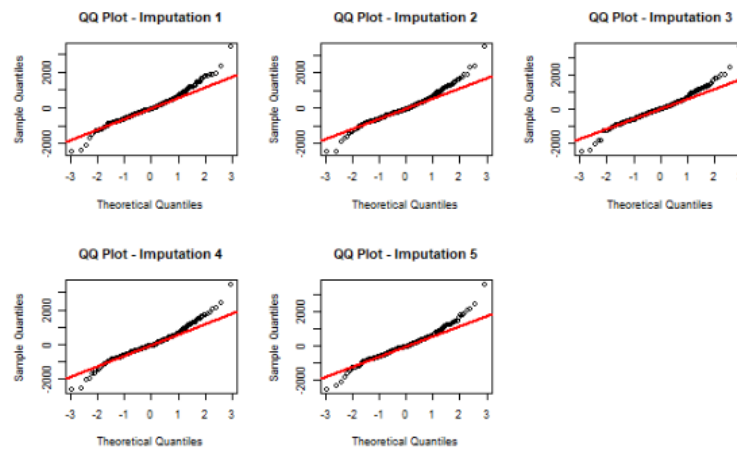
Appendix Figure 6: Residual plot for Multiple Imputation Data for SQ 1



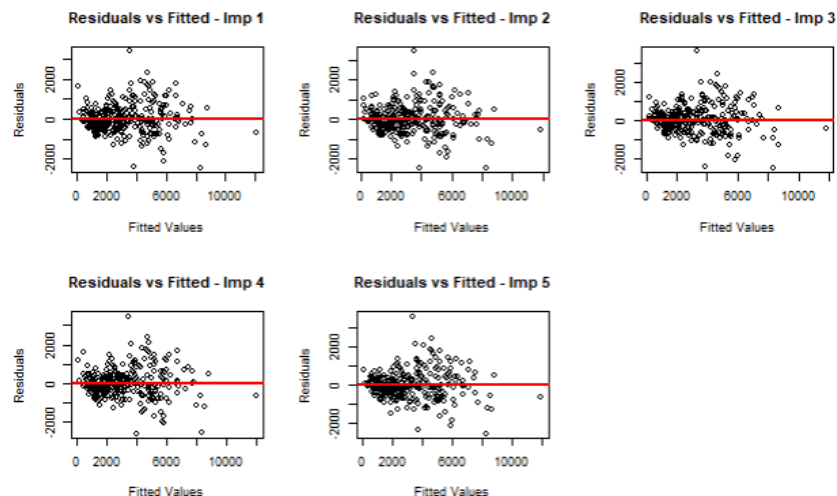
Appendix Figure 7: QQplot and residual plot for Multiple Imputation Data for SQ 2



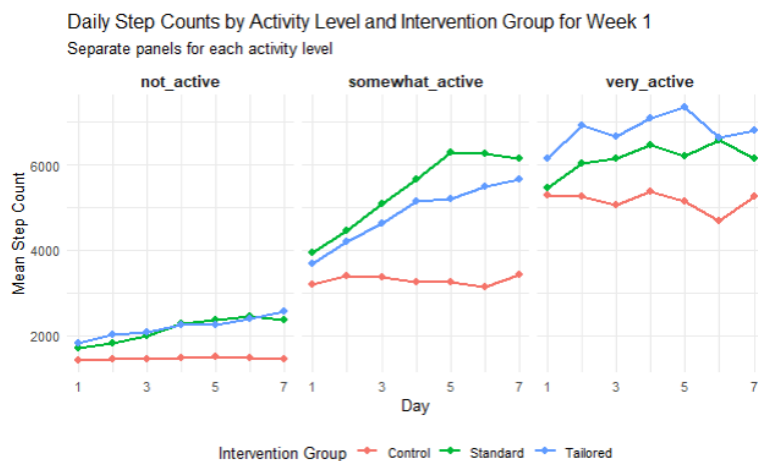
Appendix Figure 8: Log Transformed Step Count Stratified by Message Group



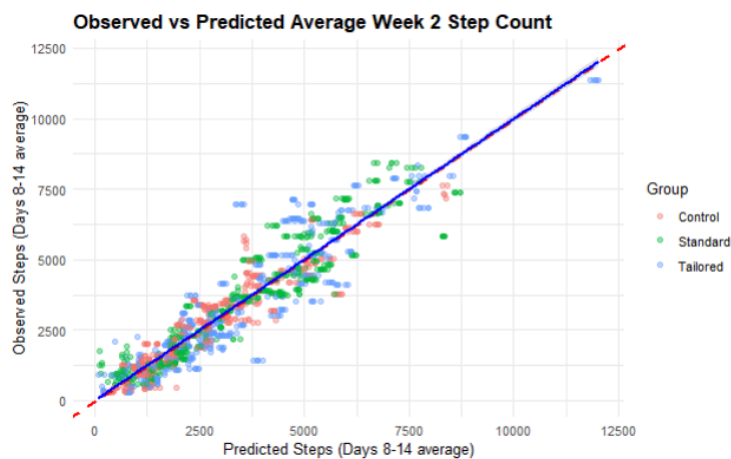
Appendix Figure 9: QQplot for Multiple Imputation Data for SQ 3



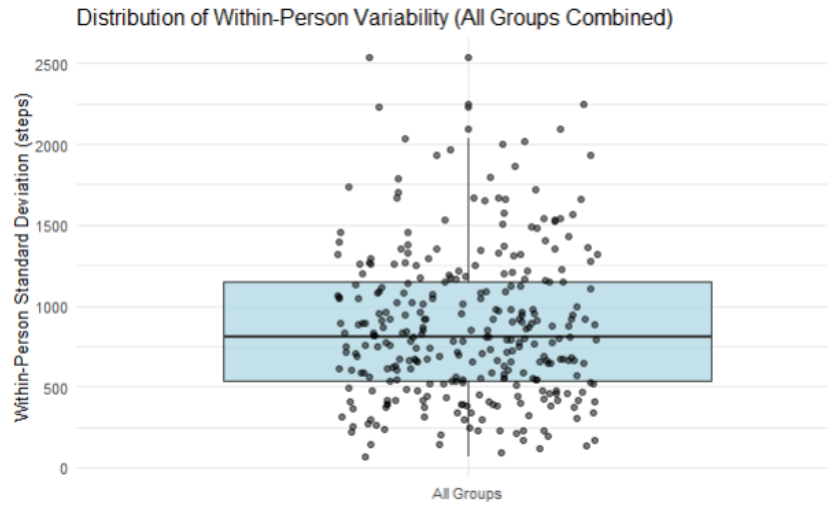
Appendix Figure 10: Residual Plot for Multiple Imputation Data for SQ 3



Appendix Figure 11: Daily Step Count by Activity Level and Intervention Group for Week 1



Appendix Figure 12: Observed vs Predicted Model of Imputed Data for Week 2



Appendix Figure 13: Distribution of Within-Person Standard Deviation
(The data points are jittered to show where subjects lie on the box plot)



Appendix Figure 14: Distribution of Within-Person Standard Deviation Stratified by Group
(The data points are jittered to show where subjects lie on the box plot)

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- [1] “Health Risks of an Inactive Lifestyle.” *MedlinePlus*, U.S. National Library of Medicine, medlineplus.gov/healthrisksofaninactivelifestyle.html. Accessed 21 June 2025.
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